TRI-OCTAGON MATHEMATICAL RESEARCH

Finite-Time Access to Native Projection Capacity in the Reciprocal Chiral Exchange Candidate

An explicit quarter-capacity certificate
for coherent empty launches

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Abstract

The joint projection ceiling of the reciprocal chiral exchange candidate limits the native trial storage that can leave the core, but a ceiling alone does not establish access to a definite fraction of that amount. We prove an explicit finite-time certificate for the coherent family $\Omega_0 = (1, e^{i\phi}, 1)$, $0 < \phi < \pi/12$, at fixed yaw $\pi/4$, with nonzero raw drive and initially empty carrier and receiver. The original equations retain finite positive propagation, finite-width periodic profiles and constant positive couplings. First fix admitted geometry and coefficients satisfying $\sigma_{\mathrm{ext}} \le \nu/2$, where $\nu^2 = \eta_N^2 m k^2 R^2/(2\alpha_N)$ and $\sigma_{\mathrm{ext}}^2 = c^2 d/m + \kappa^2 m/\beta_R$. Every phase in this family then satisfies, at the same time $t_* = 1/\nu$,

$$H_C(t_*) + H_R(t_*) > \frac{1053}{4096} C_{\mathrm{joint}} > \frac{1}{4} C_{\mathrm{joint}}, \qquad C_{\mathrm{joint}} = \alpha_N(1 - \cos\phi) > 0.$$

The proof retains the evolving launch drive in a Volterra comparison and bounds both propagation and receiver residuals of a strong reference curve. Exact rational estimates give a strict margin even at equality in the sufficient rate condition, whose finite positive regime is nonempty. The numerator is combined external stock and the denominator is joint capacity, not all initial native stock. The theorem establishes neither a receiver-only fraction nor subsequent recovery of this amount on these coefficients, saturation of the ceiling, or a spatial toroidal circuit.

Research author and direction: Hilmir Frímann Halldórsson. GPT formulated and internally reviewed the accepted investigation. Codex independently investigated the result and prepared this manuscript. Scientific acceptance is unchanged; this new manuscript and proposed publication payload await review. No external peer review or machine-formal proof is claimed.

1 From a structural ceiling to finite-time access

The reciprocal chiral exchange candidate couples a native state to a periodic carrier and a reversible receiver through two finite distributed profiles. Its original reciprocal feedback provides well-posed evolution and matching signed storage accounts [1]. A subsequent theorem establishes later native credit on a specified initially empty solution [2]. The projection-capacity analysis then supplies an additional native invariant and a sharper structural ceiling, while a separate sufficient regime yields high receiving and recovery fractions relative to an intermediate transferred stock [3]. These results distinguish how much can leave from what fraction of a particular transferred amount can return.

The remaining gap addressed here is quantitative access: does an actual unchanged solution reach a definite fraction of the joint ceiling at a finite time? A sharp minimum on an invariant level need not be reached dynamically, and high recovered fractions can coexist with small absolute transfer. For the fixed-yaw coherent empty-launch family, we establish access to more than one quarter of the joint ceiling under an explicit sufficient condition. Geometry and all coefficients are fixed before choosing the phase. The time and the fractional guarantee are uniform over the declared open phase range.

The accepted investigation [4] and its project-internal scientific review [5] are the sources of the new theorem and full proof. Section 2 fixes the model and inherited ceiling. Section 3 states the certificate; Section 4 proves it through an original-variable energy estimate, a drive-dependent Volterra bound and a reference residual. Section 5 proves nonemptiness of the finite parameter regime. Section 6 separates the resulting stock guarantee from receiving, recovery and geometric questions. Appendix A checks dimensions without assigning a physical clock.

The motivating picture retains a central starting or meeting region, opposed cones, outward spreading and a curved around-and-inward return, with stationary supports and possible transfer elsewhere. The present theorem is a storage statement for the specified candidate. A norm estimate does not assign this intended spatial route.

2 Fixed equations and inherited capacity

2.1 Registration and finite periodic profiles

Write $x = (a, b) \in \mathbb{R}^6$ and $\Omega = a + ib \in \mathbb{C}^3$. Retain the raw port order $+, -$ and the fixed rank-two registration [1, Section 1]:

$$\begin{gathered} e_y = (0,1,0), \quad e_z = (0,0,1), \quad n_\pm = (2e_y \pm \sqrt{3}e_z)/\sqrt{7}, \\ A_{\mathrm{reg}} = \lambda_g R_z(\delta_g), \quad K = \kappa_{\mathrm{in}} \begin{pmatrix} n_+^T \\ n_-^T \end{pmatrix} A_{\mathrm{reg}}, \quad G = \kappa_{\mathrm{in}}\lambda_g > 0, \\ f(x) = K(a \times b), \qquad J(x) = D_x f(x), \qquad \delta_g = \pi/4. \end{gathered} \tag{2.1}$$

Here R_z is rotation about the z axis and both registration gains are finite and positive. Contact-tangent orientation signs do not alter the raw drive.

Let $s \in \mathbb{R}/L_c\mathbb{Z}$, with finite $L_c, c > 0$. The admitted contacts $s_\pm = L_c/2 \mp \eta_{\mathrm{geom}}$ have shortest *periodic* distance $2\eta_{\mathrm{geom}}$, and $0 < \epsilon_c < \eta_{\mathrm{geom}}$. For these positions, the shortest-distance assertion includes $2\eta_{\mathrm{geom}} \le L_c/2$. This is a consequence of the stated circle hypothesis, not an additional

assumption. With $(q)_+ = \max(q, 0)$, define

$$b_\sigma(s) = \epsilon_c^{-1}\left(1 - \frac{|\operatorname{wrap}_{L_c}(s - s_\sigma)|}{\epsilon_c}\right)_+, \qquad \mathcal{B}v = b_+ v_+ + b_- v_-,$$
$$\mathcal{B}^* p = \begin{pmatrix} \int b_+ p\,\mathrm{d}s \\ \int b_- p\,\mathrm{d}s \end{pmatrix}, \qquad m = \frac{2}{3\epsilon_c}, \qquad d = \frac{2}{\epsilon_c^3}, \tag{2.2}$$
$$\mathcal{B}^*\mathcal{B} = m\mathrm{Id}_2, \qquad \|\mathcal{B}\| = \|\mathcal{B}^*\| = \sqrt{m}, \qquad \|\mathcal{B}'v\|_2 = \sqrt{d}\,|v|.$$

All integrals cover the circle. Whole-circle separation ensures disjoint support, including across the coordinate seam. The normalized tents are continuous and in H^1_{per}; their first weak derivatives lie in L^2 without corner delta terms. The identities in (2.2) are inherited from the profile calculation [1, Section 3] and its restatement [3, Section 2.1, (2.2)]. No second profile derivative is assumed.

2.2 Original exchange and signed stores

For positive finite storage scales α_N, β_R and couplings $\eta_N, \kappa = \eta_R$, the original unscaled system is

$$\dot{x} = -\frac{\eta_N}{2\alpha_N} J(x)^T y,$$
$$p_t + c(r_c)_s = \mathcal{B}\{\eta_N f(x) - \kappa z\}, \tag{2.3}$$
$$(r_c)_t + c p_s = 0, \qquad \dot{z} = \frac{\kappa}{\beta_R} y, \qquad y = \mathcal{B}^* p.$$

All coefficients remain constant along every trajectory. There is no native F jump, reset, switching or added feedback. The inherited trial stores and total signed powers are

$$H_N = \frac{\alpha_N}{2}|x|^2, \quad H_C = \frac{1}{2}(\|p\|_2^2 + \|r_c\|_2^2), \quad H_R = \frac{\beta_R}{2}|z|^2,$$
$$P_{NC} = \eta_N f(x)^T y, \qquad P_{CR} = \kappa z^T y, \tag{2.4}$$
$$\dot{H}_N = -P_{NC}, \qquad \dot{H}_C = P_{NC} - P_{CR}, \qquad \dot{H}_R = P_{CR}, \qquad H_N + H_C + H_R = \text{constant}.$$

The quadratic H_N is a trial store, distinct from the native quartic potential. Equations (2.3)–(2.4) use the original factors and signs [3, Section 2.2, (2.3)–(2.4)].

2.3 Coherent empty launch and joint ceiling

The family used throughout this paper is

$$\Omega_0 = (1, e^{i\phi}, 1), \quad 0 < \phi < \pi/12, \qquad p(0) = r_c(0) = z(0) = 0. \tag{2.5}$$

At the fixed yaw in (2.1), the published launch and projection evaluations give

$$R = |x_0| = \sqrt{3}, \qquad k = \|K\| = G\sqrt{8/7},$$
$$f_{0,\pm} = \frac{G\sin\phi}{\sqrt{7}}(-\sqrt{2} \pm \sqrt{3}), \qquad |f_0|^2 = \frac{10G^2\sin^2\phi}{7} > 0, \tag{2.6}$$
$$C_{\mathrm{joint}} = \alpha_N(1 - \cos\phi) > 0.$$

The numerical amplitudes use the published normalized native convention; Appendix A restores their dimensions. The joint ceiling combines the published bilinear and rank-two projection invariants. Its general derivation is P3 Theorem 3.1; its specialization here is P3 (3.18) [3]. Conservation and empty external preparation give

$$E_{\mathrm{out}}(t) := H_C(t) + H_R(t) = H_N(0) - H_N(t), \qquad 0 \le E_{\mathrm{out}}(t) \le C_{\mathrm{joint}} \quad (t \ge 0). \tag{2.7}$$

The drive formulas are P3 (2.7)–(2.8) specialized to this yaw. No invariance of the initial coherent phase sector or of the launch family is presumed.

3 An explicit finite-time access certificate

Define two positive estimate rates using the fixed registration, profiles and coefficients:

$$\nu^2 = \frac{\eta_N^2 m k^2 R^2}{2\alpha_N}, \qquad \sigma_{\mathrm{ext}}^2 = \frac{c^2 d}{m} + \frac{\kappa^2 m}{\beta_R}. \tag{3.1}$$

Theorem 3.1 (Quarter-capacity access)**.** *First fix admitted geometry and all finite positive coefficients of Section 2, at yaw $\pi/4$, satisfying*

$$\sigma_{\mathrm{ext}} \le \frac{\nu}{2}, \qquad t_* = \frac{1}{\nu}. \tag{3.2}$$

For every $0 < \phi < \pi/12$, the unique unchanged solution of (2.3) with the empty launch (2.5) then satisfies

$$\boxed{H_C(t_*) + H_R(t_*) > \frac{1053}{4096} C_{\mathrm{joint}} > \frac{1}{4} C_{\mathrm{joint}} > 0.} \tag{3.3}$$

The same coefficients and the same phase-independent time t_* apply throughout this family. Equality in the sufficient rate condition is permitted.

The proof occupies Section 4. Section 5 shows that the allowed finite positive regime is nonempty. The theorem concerns combined external stock at one specified time relative to the joint ceiling. The rates in (3.1) are analytical estimate scales; neither identifies a physical native frequency or native-map cadence. No necessary or optimal threshold is asserted.

4 Proof of the finite-time certificate

4.1 Inherited strong evolution and the original forced estimate

P1 Theorem 3.1 is the global exchange theorem [1]; it supplies existence and uniqueness before any estimate below. It is distinct from P2 Theorem 3.1, the separate native-credit theorem [2]. For the empty H^1 carrier preparation, the actual solution satisfies

$$\begin{aligned} &x, z \in C^1(\mathbb{R}), \qquad (p, r_c) \in C(\mathbb{R}; H^1_{\mathrm{per}})^2 \cap C^1(\mathbb{R}; L^2)^2, \\ &|x(t)| \le R, \qquad \|J(x(t))\| \le kR, \qquad |y(t)| \le \sqrt{m}\, \|U(t)\|_{CR}, \\ &U = (p, r_c, z), \qquad \|U\|_{CR}^2 = \|p\|_2^2 + \|r_c\|_2^2 + \beta_R |z|^2 = 2E_{\mathrm{out}}. \end{aligned} \tag{4.1}$$

Conservation and nonnegative external stores give the native ball. For $h = (h_a, h_b)$, $J(x)h = K(h_a \times b + a \times h_b)$ and $|h_a||b| + |a||h_b| \le |x||h|$ give $\|J(x)\| \le k|x|$. The port bound follows from (2.2) and the carrier component of the weighted norm. These are the unscaled bounds recorded in P3 (5.5) [3].

On $\mathcal{H}_{\mathrm{ext}} = L^2_{\mathrm{per}} \times L^2_{\mathrm{per}} \times \mathbb{R}^2$ use the weighted inner product of (4.1). Define

$$\begin{aligned} \mathcal{L}U &= \left(-c(r_c)_s - \kappa \mathcal{B} z,\ -c p_s,\ \frac{\kappa}{\beta_R} \mathcal{B}^* p \right), \\ D(\mathcal{L}) &= H^1_{\mathrm{per}} \times H^1_{\mathrm{per}} \times \mathbb{R}^2, \qquad \dot{U} = \mathcal{L}U + (\eta_N \mathcal{B} f(x), 0, 0). \end{aligned} \tag{4.2}$$

For $V = (q_c, h, v) \in D(\mathcal{L})$, periodic Sobolev integration and adjointness give

$$\langle V, \mathcal{L}V \rangle_{CR} = -c \int [q_c h_s + h(q_c)_s]\, \mathrm{d}s - \kappa \langle q_c, \mathcal{B} v \rangle_2 + \kappa v \cdot \mathcal{B}^* q_c = 0. \tag{4.3}$$

The carrier integrand is the weak derivative of the periodic product $q_c h$, whose integral vanishes. The receiver weight β_R cancels the denominator in (4.2), leaving two equal and opposite κ terms.

For a known strong curve $V \in C([0,T];D(\mathcal{L})) \cap C^1([0,T];\mathcal{H}_{\mathrm{ext}})$ with continuous forcing F, this yields

$$\dot{V} = \mathcal{L}V + F \quad \Longrightarrow \quad \|V(t)\|_{CR} \le \|V(0)\|_{CR} + \int_0^t \|F(s)\|_{CR}\,\mathrm{d}s. \tag{4.4}$$

For completeness, first regularize the norm with $\varepsilon > 0$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\sqrt{\|V(t)\|_{CR}^2 + \varepsilon^2} = \frac{\langle V(t), F(t)\rangle_{CR}}{\sqrt{\|V(t)\|_{CR}^2 + \varepsilon^2}} \le \|F(t)\|_{CR}\,. \tag{4.5}$$

Integrate and take $\varepsilon \downarrow 0$ to prove (4.4), including at zeros. No unregularized norm is differentiated at zero. Existence and strong regularity are inherited, not inferred from cancellation.

4.2 A drive-dependent Volterra comparison

Set $A_0 = \eta_N\sqrt{m}\,|f_0| > 0$, a forcing magnitude distinct from the published drift constant and accessibility statistic. Applying (4.4) to the actual empty state and using the finite-dimensional chain rule gives

$$\begin{aligned} \|U(t)\|_{CR} &\le \eta_N\sqrt{m}\int_0^t |f(x(s))|\,\mathrm{d}s, \\ \dot{f}(x(t)) &= J(x(t))\dot{x}(t) = -\frac{\eta_N}{2\alpha_N} J(x(t))J(x(t))^T y(t), \\ |f(t) - f_0| &\le \frac{\eta_N k^2 R^2\sqrt{m}}{2\alpha_N}\int_0^t \|U(s)\|_{CR}\,\mathrm{d}s. \end{aligned} \tag{4.6}$$

The chain rule is legitimate since x is C^1 and f is quadratic; the last bound uses $\left\|JJ^T\right\| \le k^2R^2$ and (4.1).

For the continuous nonnegative functions $g = \|U\|_{CR}/A_0$ and $h = |f - f_0|/|f_0|$, we obtain

$$\begin{aligned} g(t) &\le t + \int_0^t h(s)\,\mathrm{d}s, \qquad h(t) \le \nu^2\int_0^t g(s)\,\mathrm{d}s, \\ g(t) &\le t + (\mathcal{V}g)(t), \qquad (\mathcal{V}v)(t) = \nu^2\int_0^t (t-s)v(s)\,\mathrm{d}s. \end{aligned} \tag{4.7}$$

Indeed $\eta_N k^2R^2\sqrt{m}\,A_0/(2\alpha_N|f_0|) = \nu^2$, retaining both native-coupling factors. Interchanging the continuous nonnegative double integral yields the last inequality.

On each finite $[0,T]$ the operator $\mathcal{V}$ preserves order. Iteration gives

$$\begin{aligned} g(t) &\le \sum_{j=0}^{n-1}\frac{\nu^{2j}t^{2j+1}}{(2j+1)!} + (\mathcal{V}^n g)(t), \\ 0 \le (\mathcal{V}^n g)(t) &\le M_T\frac{\nu^{2n}t^{2n}}{(2n)!} \le M_T\frac{\nu^{2n}T^{2n}}{(2n)!} \longrightarrow 0, \qquad M_T = \sup_{0\le s\le T} g(s) < \infty. \end{aligned} \tag{4.8}$$

The supporting monomial identity and successive remainder ratio are

$$\int_0^t (t-s)s^\ell\,\mathrm{d}s = \frac{t^{\ell+2}}{(\ell+1)(\ell+2)}, \qquad \frac{(\nu T)^2}{(2n+2)(2n+1)} \longrightarrow 0. \tag{4.9}$$

The identity proves the summands and the remainder bound by induction from $\mathcal{V}^0 t = t$ and $\mathcal{V}^0 1 = 1$. Continuity gives finite M_T, and the ratio proves uniform disappearance of the remainder on the finite interval. No differential inequality for g at zero is assumed. Summing the series and substituting into the bound for h give, for every finite $t \ge 0$,

$$\boxed{\|U(t)\|_{CR} \le \frac{A_0}{\nu}\sinh(\nu t), \qquad |f(t) - f_0| \le |f_0|[\cosh(\nu t) - 1].} \tag{4.10}$$

Both estimates retain the actual launch-drive factor. The iteration applies separately at every admitted phase and requires no phase-uniform preliminary bound on M_T; its final formula applies to all those phases.

4.3 Strong reference, full residual and finite error

Use the comparison curve

$$U_{\mathrm{ref}}(t) = (\eta_N t\mathcal{B}f_0, 0, 0), \qquad \|U_{\mathrm{ref}}(t)\|_{CR} = A_0 t, \qquad W = U - U_{\mathrm{ref}}. \tag{4.11}$$

Since $\mathcal{B}f_0 \in H^1_{\mathrm{per}}$, the reference is C^1 with values in $D(\mathcal{L})$. The actual state already has (4.1). Thus the difference is strong *before* its residual is estimated: its carrier lies in $C_t H^1_{\mathrm{per}} \cap C^1_t L^2$ and its receiver is C^1. This retains the regularity order used in P3 Section 5.2 [3]. It does not define a replacement dynamical system.

Substitute $U = W + U_{\mathrm{ref}}$ into (4.2) and subtract $\dot{U}_{\mathrm{ref}}$:

$$\begin{aligned} \dot{W} &= \mathcal{L}W + (\eta_N\mathcal{B}(f - f_0), 0, 0) + F_{\mathrm{ref}}(t), \qquad W(0) = 0, \\ F_{\mathrm{ref}}(t) &= \mathcal{L}U_{\mathrm{ref}}(t) = \left(0, -c\eta_N t\mathcal{B}'f_0, \frac{\kappa\eta_N m t}{\beta_R} f_0\right). \end{aligned} \tag{4.12}$$

The first component vanishes because the reference has zero r_c, z. Its second component is the negative propagation term $-c\partial_s p_{\mathrm{ref}}$. Its positive receiver term is $\kappa\mathcal{B}^* p_{\mathrm{ref}}/\beta_R$, using the exact Gram matrix $m\mathrm{Id}_2$. That term is nonzero despite the reference receiver being zero. Both actual c and actual κ remain positive.

These two residual components are orthogonal in the weighted space, so

$$\begin{aligned} \|F_{\mathrm{ref}}(t)\|^2_{CR} &= \eta_N^2 |f_0|^2 t^2 \left(c^2 d + \frac{\kappa^2 m^2}{\beta_R}\right) = A_0^2 \sigma_{\mathrm{ext}}^2 t^2, \\ \|F_{\mathrm{ref}}(t)\|_{CR} &= A_0 \sigma_{\mathrm{ext}} t \qquad (t \geq 0). \end{aligned} \tag{4.13}$$

Only $\mathcal{B}'f_0 \in L^2$ is needed. The residual need not be H^1, and it is not differentiated spatially again. Strong regularity is not inferred from this residual; no second tent derivative or operator-norm limit of an unbounded propagation operator is used.

Apply (4.4) to the already strong difference W and use (4.10) and (4.13):

$$\begin{aligned} \|U(t) - U_{\mathrm{ref}}(t)\|_{CR} &\leq \eta_N \sqrt{m} \int_0^t |f(s) - f_0| \, \mathrm{d}s + A_0 \sigma_{\mathrm{ext}} \int_0^t s \, \mathrm{d}s \\ &\leq A_0 \left[\frac{\sinh(\nu t)}{\nu} - t + \frac{\sigma_{\mathrm{ext}} t^2}{2}\right]. \end{aligned} \tag{4.14}$$

This finite-time error includes native drift and the complete external residual, and is valid before imposing (3.2).

4.4 Positive bracket and the exact strict margin

Set $u = \nu t$ and $\rho = \sigma_{\mathrm{ext}}/\nu$. Reverse triangle gives

$$\|U(t)\|_{CR} \geq \frac{A_0}{\nu} B(u, \rho), \qquad B(u, \rho) = 2u - \sinh u - \frac{\rho u^2}{2}. \tag{4.15}$$

If the right side is negative it remains a lower bound, but it cannot be squared to obtain a stock bound. *Only when* $B(u, \rho) > 0$, the positive capacity and the original stock factor $\|U\|^2_{CR}/2$ give

$$\frac{E_{\mathrm{out}}(t)}{C_{\mathrm{joint}}} \geq \frac{\alpha_N |f_0|^2}{k^2 R^2 C_{\mathrm{joint}}} B(u, \rho)^2, \qquad \frac{A_0^2}{2\nu^2} = \frac{\alpha_N |f_0|^2}{k^2 R^2}. \tag{4.16}$$

At $u = 1$ and $\rho \le 1/2$, $B(1,\rho) \ge 7/4 - \sinh 1$. To certify positivity exactly, write $a_j = 1/(2j+1)!$. For $j \ge 1$,

$$\frac{a_{j+1}}{a_j} = \frac{1}{(2j+2)(2j+3)} \le \frac{1}{20}. \tag{4.17}$$

Equality holds at $j = 1$ and all later ratios are strictly smaller. The positive tail starting with $a_1 = 1/6$ is therefore strictly below its geometric majorant:

$$\sinh 1 < 1 + \frac{1/6}{1 - 1/20} = \frac{67}{57} < \frac{19}{16}, \qquad \frac{19}{16} - \frac{67}{57} = \frac{11}{912} > 0. \tag{4.18}$$

Consequently

$$B(1,\rho) \ge \frac{7}{4} - \sinh 1 > \frac{7}{4} - \frac{19}{16} = \frac{9}{16} > 0. \tag{4.19}$$

This licenses (4.16), also at equality in the rate condition.

The capacity factor is exactly

$$\frac{\alpha_N |f_0|^2}{k^2 R^2 C_{\text{joint}}} = \frac{5}{12}\frac{\sin^2\phi}{1-\cos\phi} = \frac{5}{12}(1+\cos\phi) > \frac{13}{16}. \tag{4.20}$$

To justify the strict inequality without a decimal approximation, use

$$\begin{aligned} 3 - \left(\frac{161}{100}\right)^2 &= \frac{4079}{10000} > 0, \\ \cos^2(\pi/12) = \frac{2+\sqrt{3}}{4} &> \frac{2 + 161/100}{4} = \left(\frac{19}{20}\right)^2. \end{aligned} \tag{4.21}$$

Positive square roots give $\cos(\pi/12) > 19/20$. Cosine is strictly decreasing on $[0, \pi/12]$, so $\cos\phi > 19/20$ for every admitted phase, and $(5/12)(1 + 19/20) = 13/16$. Division by $1 - \cos\phi$ in (4.20) is legitimate because $\phi > 0$.

Combining (4.16), (4.19) and (4.20) proves

$$\frac{E_{\text{out}}(1/\nu)}{C_{\text{joint}}} > \frac{13}{16}\left(\frac{9}{16}\right)^2 = \frac{1053}{4096} = \frac{1}{4} + \frac{29}{4096} > \frac{1}{4}. \tag{4.22}$$

Multiplication by $C_{\text{joint}} > 0$ proves Theorem 3.1. The certificate is finite and exact: neither a computed trajectory nor an asymptotic remainder enters the conclusion. □

5 A nonempty finite positive regime

Both rates are positive, so squaring the condition and multiplying by the positive factor $8\alpha_N/(mk^2R^2)$ give the equivalence

$$\sigma_{\text{ext}} \le \frac{\nu}{2} \quad \Longleftrightarrow \quad \eta_N^2 \ge \frac{8\alpha_N}{mk^2R^2}\left(\frac{c^2 d}{m} + \frac{\kappa^2 m}{\beta_R}\right). \tag{5.1}$$

For any fixed admitted geometry and fixed finite positive $c, \kappa, \alpha_N, \beta_R, G$, the right side is finite and positive. Select η_N as its positive square root, or any larger finite number, once before choosing the phase. Keep that value constant on every resulting solution. This gives a nonempty regime with genuinely positive propagation and receiver coupling, including the equality case.

The normalized $R^2 = 3$ is independent of phase, and k, m, d depend only on the fixed registration and profiles. Hence both (5.1) and $t_* = 1/\nu$ are phase independent. The theorem compares separately prepared trajectories under the same coefficients; it does not select coefficients along a trajectory. No zero-coupling limit, infinite-coupling construction or threshold optimization is needed. Appendix A verifies that both rates have units of inverse time and the threshold for η_N^2 is dimensionless, retaining R^2 in every dimensional formula.

6 Meaning, evidence and continuing questions

6.1 What the accessible amount measures

The numerator in (3.3) is the combined carrier-plus-receiver trial stock at one specified time. Its denominator is joint structural capacity. It is not a claim to extract one quarter of all initial native stock: in this normalized family,

$$\frac{C_{\mathrm{joint}}}{H_N(0)} = \frac{\alpha_N(1-\cos\phi)}{3\alpha_N/2} = \frac{2}{3}(1-\cos\phi). \tag{6.1}$$

The capacity-normalized guarantee is uniform over the open phase interval, while the absolute stock tends to zero as $\phi \downarrow 0$: $0 \le E_{\mathrm{out}}(t_*) \le \alpha_N(1-\cos\phi) \to 0$ by (2.7). The zero-drive endpoint is excluded and is never used as a zero denominator. The empty-launch account also retains $H_N(t) \le H_N(0)$.

6.2 Receiving and recovery retain their own hypotheses

P3's accessibility statistic $\mathcal{A}_\kappa = M_\kappa/C_{\mathrm{inv}}$ retains its original denominator and compact time window [3, Section 6.1, (6.1)–(6.2)]. The present paper states a direct inequality for $E_{\mathrm{out}}(1/\nu)/C_{\mathrm{joint}}$ without reassigning that symbol. P3's distinction between absolute amount and recovered fraction remains in force [3, Section 6.2, (6.3)–(6.5)].

Neither a receiver-only share nor later recovery of this particular capacity-sized amount is established by Theorem 3.1. P3's earlier 80/81 receiving and net-recovery guarantees and P2's native-credit theorem retain their own sufficient hypotheses [3, Section 4], [2, Theorem 3.1]. Their overlap with (5.1) on one unchanged trajectory has not been proved here. Those fractions and return conclusions are therefore not attached to the new amount. No ceiling saturation, maximizing time, monotone extraction, physical-energy calibration or completed spatial circuit follows.

6.3 Evidence and contribution status

The accepted proof and current internal scientific review are preserved unchanged with the candidate [4, 5]. The written argument carries strong regularity, zero-norm regularization, positive-kernel order, disappearance of the factorial remainder, strictness and quantifiers. The original exact-algebra evidence has eight focused groups; the subsequent independent review replayed those same eight groups. They are not sixteen distinct scientific discoveries, and software checking does not replace the analytical arguments.

This manuscript preparation performs document compilation, source/fidelity checks and all-page inspection. It does not rerun scientific checks, prior simulations or parameter scans. Hashes establish byte identities and preservation, not mathematical truth. GPT formulated and internally reviewed the investigation; Codex independently investigated it and prepared the present manuscript under the research direction of Hilmir Frímann Halldórsson. The mathematics is accepted unchanged. This newly prepared manuscript and proposed publication payload remain pending review; project-internal review is neither external peer review nor machine-formal verification.

6.4 The geometric motivation and next question

The intended central starting or meeting region, opposed cones, outward spreading, curved around-and-inward return, stationary supports and possible transfer elsewhere remain the motivation. The periodic carrier circle is distinct from the intended tight toroid. The close original enclosing sphere and the extended-line sphere are distinct, as are finite host-notch N regions and central broken G boundaries. This estimate assigns no spatial route, physical receiving location, opening or diversion coefficient. The author's tentative 15–20%, possibly more, remains geometric intuition, not a calibrated fraction or the denominator of this certificate.

The next substantive question is receiving and recovering the capacity-sized transferred amount on the same fixed-parameter solution. It is not investigated here. The standing sequence then turns to native F/exchange cadence, intended-route correspondence and six-opening transport, with linked gravity–magnetism, later outer geometry and justified SRG integration. Three-Way remains parked. Brainstorming material does not supply a new premise or theorem for this manuscript.

7 Conclusion

For the specified coherent empty-launch family at yaw $\pi/4$, one fixed finite positive coefficient choice satisfying $\sigma_{\mathrm{ext}} \leq \nu/2$ guarantees combined external stock strictly exceeding $1053/4096$ of joint capacity at the phase-independent time $1/\nu$. The proof controls the actual evolving drive and both components of the external reference residual, with exact strict rational margins. The nonempty sufficient regime requires no change to the original equations or coefficients along a trajectory.

This advances the joint ceiling from a necessary bound to a definite finite-time access guarantee in the declared regime. Receiving and recovering that capacity-sized amount on the same coefficients, as well as the intended spatial route, remain subsequent questions. The accepted mathematics is preserved unchanged; the new manuscript and publication candidate await explicit manuscript-and-payload review.

A Dimensions and normalized amplitudes

The dimensional conventions follow P1 Section 1, Table 2 and P3 Appendix A, Table 1 [1, 3]. Let U denote carrier amplitude, L length, T exchange time and $\mathrm{A}_{\mathrm{amp}}$ native amplitude. The numerical launch identities in (2.6) use one native amplitude unit; restoring it attaches the corresponding amplitude powers to those identities. In particular, R has an amplitude dimension even though its normalized value is $\sqrt{3}$.

Table 1: Dimensions in the original exchange variables.

Quantity	Dimension
x, R	$\mathrm{A}_{\mathrm{amp}}$
k, K	$\mathrm{UL}/(\mathrm{TA}_{\mathrm{amp}}^2)$
J	$\mathrm{UL}/(\mathrm{TA}_{\mathrm{amp}})$
p, r_c, y	U
f, z	UL/T
α_N	$\mathrm{U}^2\mathrm{L}/\mathrm{A}_{\mathrm{amp}}^2$
β_R	T^2/L
η_N, κ	1
m, d, c	$\mathrm{L}^{-1},\ \mathrm{L}^{-3},\ \mathrm{L/T}$
$\|U\|_{CR}, A_0$	$\mathrm{UL}^{1/2},\ \mathrm{UL}^{1/2}/\mathrm{T}$
$\nu, \sigma_{\mathrm{ext}}, t_*$	$\mathrm{T}^{-1},\ \mathrm{T}^{-1},\ \mathrm{T}$
$E_{\mathrm{out}}, C_{\mathrm{joint}}$	$\mathrm{U}^2\mathrm{L}$

Explicitly, the cancellations required by (3.1) are

$$\left[\frac{mk^2R^2}{\alpha_N}\right] = \mathrm{T}^{-2}, \qquad \left[\frac{c^2d}{m}\right] = \mathrm{T}^{-2}, \qquad \left[\frac{\kappa^2 m}{\beta_R}\right] = \mathrm{T}^{-2}. \tag{A.1}$$

Thus $B(u, \rho)$ is dimensionless, A_0/ν has the external-norm dimension, and the right side of (5.1) is dimensionless like η_N^2. Retaining R^2 is essential. These are commensurability checks, not physical calibration, native-frequency identification, an F-step cadence or evidence of a completed carrier circulation.

References

[1] H. Frímann Halldórsson, *A Reciprocal Chiral Exchange Extension of the Tri-Octagon Model*, v0.1 (2026). Immutable published edition and evidence. Theorem 3.1 supplies global well-posedness and strong empty-launch regularity.

[2] H. Frímann Halldórsson, *Same-Trajectory Native Credit in the Reciprocal Chiral Exchange Candidate*, v0.1 (2026). Immutable published edition and evidence. Theorem 3.1 is the separate native-credit theorem.

[3] H. Frímann Halldórsson, *Recoverable Transfer and Projection Capacity in the Reciprocal Chiral Exchange Candidate*, v0.1 (2026). Immutable published edition and evidence. Theorem 3.1 supplies the projection ceiling; Section 4 supplies the separate receiving/recovery regime.

[4] H. Frímann Halldórsson, with GPT formulation and Codex independent investigation, *Native finite capacity access: independent proof assessment*, 4 November 2026. Accepted proof, claim/source ledger and original eight-group exact-algebra evidence, preserved with this manuscript.

[5] GPT, *Review of Native Finite Capacity Access 01*, 4 November 2026. Project-internal scientific acceptance unchanged; independent analytical review and replay of the same eight algebra groups. This review accepts the investigation, not the present new manuscript.